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1. Introduction

Survey Corps Global Retail (SCGR) is a multi-seller e-commerce platform that connects independent sellers with customers across different regions in Brazil. The company offers a wide range of products, including electronics, fashion, and homeware. Operating in a highly competitive marketplace, SCGR's performance relies heavily on its ability to effectively manage, interpret, and make informed decisions from the large volumes of data generated through sales, deliveries, customer reviews, and seller activities (Sharda et al., 2020). To study the organisation's performance challenges, this report uses the Brazilian E-Commerce Public Dataset from Kaggle (2021).

After an initial review of the dataset, several issues with potential implications for SCGR's overall performance were identified. One major area relates to delivery delays, with around 6.8% of orders arriving later than the promised date. Delivery performance is closely linked to customer satisfaction (Ramanathan, 2010), and this pattern was reflected in the review data. Although the average review score is around 4.09, a sizeable number of low ratings were associated with late deliveries, damaged items or unmet expectations (Filieri, 2015). There is also variation across categories and states in sales volumes and review patterns. These inconsistencies indicate possible weaknesses in SCGR's demand forecasting, inventory planning and logistics coordination. Currently, SCGR lacks a unified system that integrates these diverse perspectives to facilitate rapid and informed decision-making.

This lack of an integrated Business Intelligence (BI) environment creates several challenges. For example, managers currently do not have a single view that combines key indicators like sales, delivery performance, and regional demand. Instead, they have a system where essential information is scattered across different platforms, making it difficult to understand where problems occur or how different parts of the business interact. This fragmentation prevents SCGR from seeing the connections between sales, delivery performance and customer satisfaction (Otto, 2011). This can lead to reactive decision-making, slower responses to issues and missed opportunities for improvement. Therefore, as SCGR continues to grow, these limitations could harm its competitiveness, efficiency and customer trust.

The purpose of this report is to demonstrate how a structured BI approach can help SCGR improve visibility over its operations and address the performance challenges identified. The analysis follows the Data Analytics Lifecycle and uses a dimensional model to support a Power BI dashboard built from the Kaggle dataset (2021). The dashboard integrates sales, customer, logistics and review data to provide a clearer view of SCGR's operations.

2. Theoretical Framework

SCGR's main challenge is that operational data exists in disconnected parts, making it difficult for decision-makers to form a complete view of performance. Sharda et al. (2020) emphasise that reliable and integrated information is essential for effective decision-making, and when data is fragmented, businesses often struggle to identify issues or plan strategically.

From a business analysis perspective, organisations must ensure that their information flows support their broader strategic objectives. As highlighted by Paul et al. (2014), incomplete or isolated information makes decision-making weaker, as managers cannot link actions to outcomes. This is particularly significant for SCGR as it needs to connect sales patterns, customer behaviour, logistics performance and seller reliability, to understand what drives its operational issues. The current lack of integration prevents the company from aligning its processes with its goals, leading to inefficiencies in forecasting, customer service and seller management.

To address these challenges, the implementation of a data analytics solution is not merely a technical upgrade but a strategic imperative. The literature on Big Data emphasises that the ability to process and analyse large volumes of data is a key driver of competitive advantage. Kopanakis et al. (2016) show that firms using data-driven innovation improve their competitive advantage, respond better to market changes and develop more effective strategies. Similarly, Müller et al. (2018) provide evidence that organisations that adopt strong analytics capabilities experience better overall performance and productivity. For SCGR that operates in a data-rich and rapidly changing environment, adopting a BI approach is not only beneficial but essential for long-term growth.

This project followed the Data Analytics Lifecycle to ensure rigour and systematically provide solutions for SCGR's challenges. The lifecycle began with the discovery phase, where the Brazilian E-Commerce Public Dataset (Kaggle, 2021) was selected as the empirical basis for the analysis. The dataset was considered appropriate because it covers the entire chain of SCGR's operations, including customers, sellers, products, payments, order items, delivery timestamps and review scores.

Following discovery, the data preparation phase addressed the quality issues inherent in raw datasets by focusing on cleaning and organising the data to ensure accuracy. In line with Sharda et al.'s (2020) emphasis on data preparation and data quality, the data cleaning process involved handling missing values, standardising date formats, filtering incomplete entries and merging tables using key identifiers such as 'customer_id' and 'seller_id'. Additional fields, such as 'delivery_delay_days' and 'total_order_value', were created to support deeper analysis.

The final stage of the lifecycle involved designing a data model, planning and building visualisations that could support SCGR's analytical needs. A Star Schema approach was utilised, a standard architecture in Business Intelligence design. This model simplifies complex data relationships and enhances performance when creating dashboards. In this model, transactional tables were organised as facts and supported by descriptive dimensions, enabling a clear analytical structure aligned with Kimball's dimensional modelling principles. Kimball and Ross (2013) emphasise that dimensional modelling is the most effective approach for BI systems because it reduces redundancy and enables multi-dimensional analysis through filters and drill-downs.

The full structure of the model is shown in the ERD figure below:

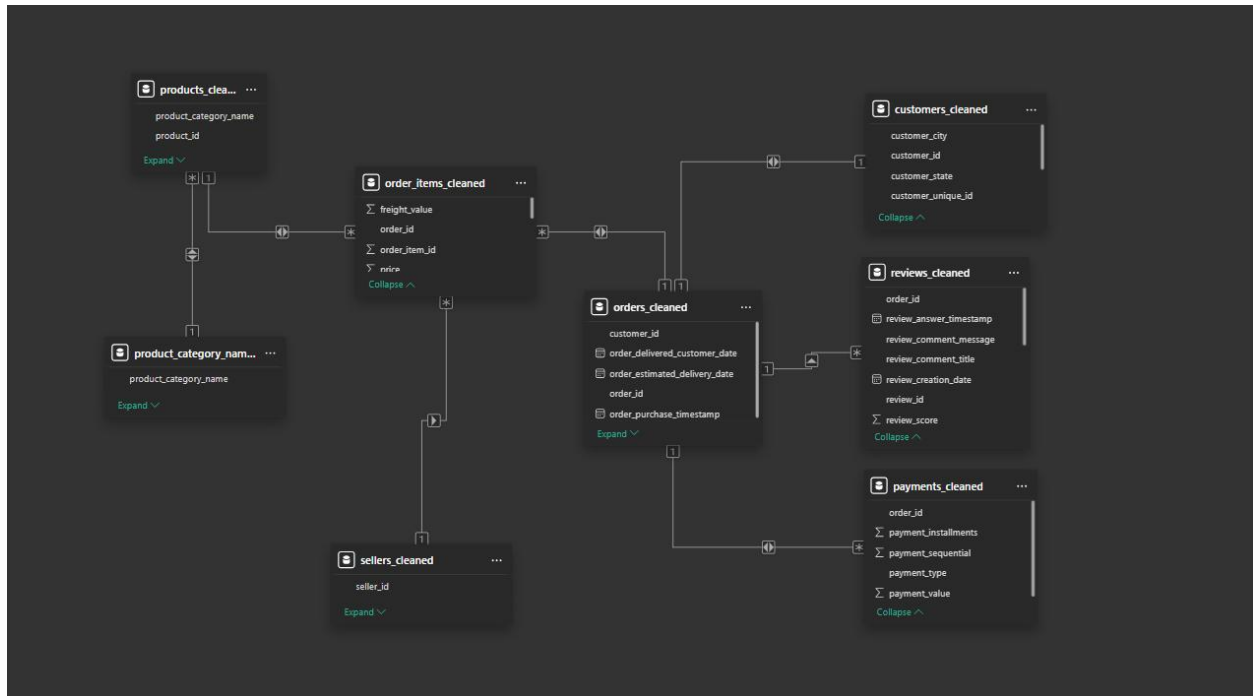


Figure 1: Entity–Relationship Diagram for SCGR’s Analytical Data Model (Author’s own work, 2025)

The ERD also aligns with established BI system architecture. Building on Eckerson’s (2010) description of BI architectures, BI systems can be viewed as operating across three layers: a data layer that integrates and stores information, an analytics layer where calculations and models are applied, and a presentation layer that displays insights through dashboards. By consolidating orders, customer data, logistics information and reviews into a single analytical environment, SCGR’s model follows this structure. This integrated foundation ensures that insights generated through the dashboard are consistent and relevant to the organisation’s decision-making needs.

Together, these theoretical foundations support the development of the dashboard presented in the next section of the report, which consolidates SCGR’s fragmented data into a unified analytical view, providing clearer insight into its operational performance.

3. Dashboard Insights and Interpretation

The Business Intelligence dashboard developed for SCGR integrates sales, logistics and customer experience data to provide a consolidated view of performance across the organisation. By combining KPIs, analytical visualisations and interactive filters, the dashboard enables managers to explore operational issues, particularly late deliveries and inconsistent customer satisfaction, through a structured and intuitive analytical environment. The following subsections justify each component of the dashboard and explain how they support managerial decision-making.

3.1 Dashboard Overview

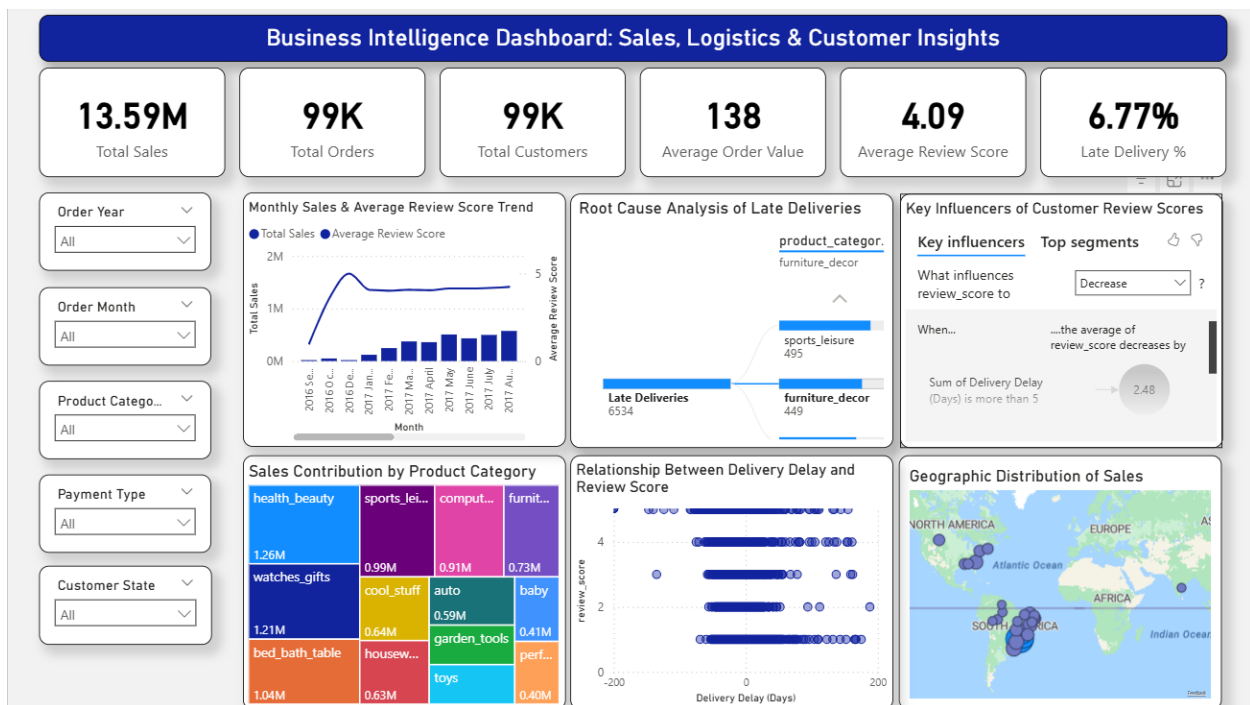


Figure 2: SCGR Business Intelligence Dashboard Overview

Figure 2 above presents the complete view of the dashboard interface. It brings together the main sales, logistics, and customer review indicators relevant to SCGR's performance. The layout prioritises clarity by placing high-level KPIs at the top, followed by analytical charts that explain trends, uncover root causes and highlight relationships between variables. This unified design

ensures that managers can move from broad performance monitoring to deeper diagnosis within a single view, which directly addresses SCGR's earlier challenge of fragmented information.

3.2 KPI Cards



Figure 3: KPI Cards

The KPI cards summarise the most critical performance indicators: **Total Sales (13.59M)**, **Total Orders (99K)**, **Total Customers (99K)**, **Average Order Value (138)**, **Average Review Score (4.09)** and **Late Delivery % (6.77%)**. These metrics provide managers with an immediate overview of the organisation's scale, revenue performance and customer satisfaction levels. One of SCGR's main operational issues, which is the Late Delivery rate, is clearly displayed at the top of the dashboard. This placement ensures that decision-makers remain aware of the significance of delivery reliability and can monitor improvements over time.

3.3 Interactive Filters (Slicers)

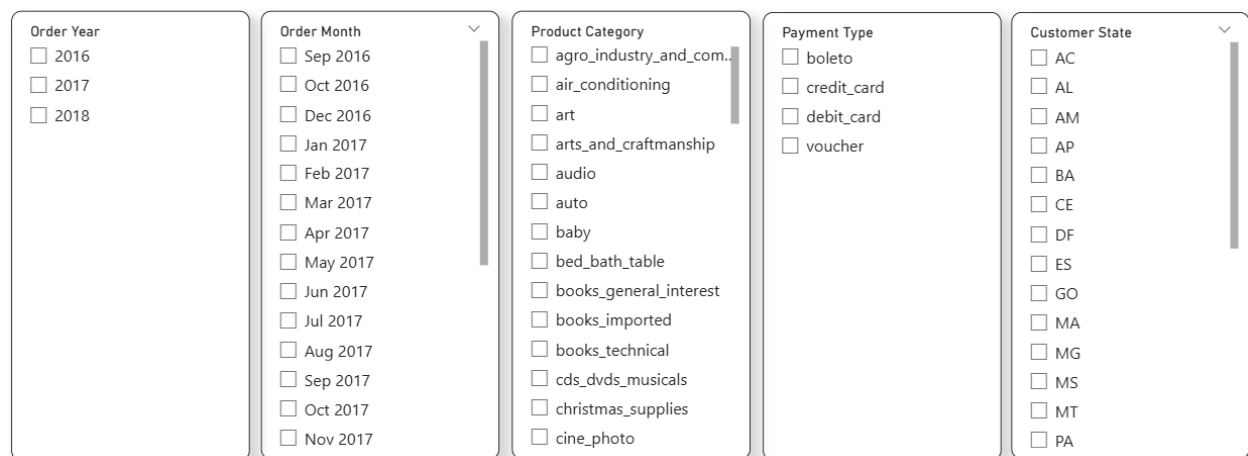


Figure 4: Dashboard Slicers

The slicers allow users to filter the dashboard by **Order Year**, **Order Month**, **Product Category**, **Payment Type**, and **Customer State**, enabling managers to isolate specific segments

of the business and monitor how performance varies across time periods, regions or product categories. The analytical value of the dashboard increases as patterns such as state-level delivery delays or category-specific satisfaction trends become easier to detect through this interactivity. For SCGR, the slicers are essential for identifying where operational inconsistencies occur.

3.4 Monthly Sales and Average Review Score Trend

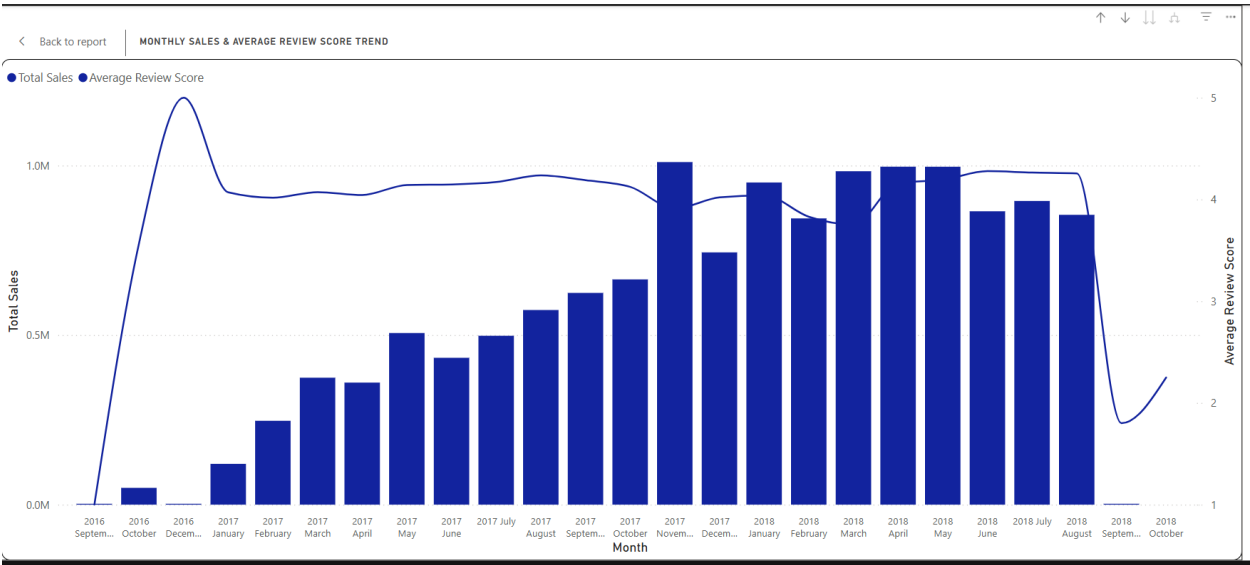


Figure 5: *Monthly Sales and Average Review Score Trend*

Figure 5 provides a dual perspective on performance by tracking monthly sales against customer satisfaction. The bar chart shows an overall growth in sales despite minor fluctuations from late 2016 through 2018, reflecting increasing platform activity and strong overall demand. However, this increase in volume did not negatively impact service quality, as review scores remained relatively stable around 4.0–4.2 for most months. Although both metrics dipped significantly in September 2018, this may indicate a temporary operational issue or incomplete data reporting. Managers can use this insight to align sales growth with service quality improvements, ensuring that customer satisfaction does not decline as order volume increases.

3.5 Root Cause Analysis of Late Deliveries (Decomposition Tree)

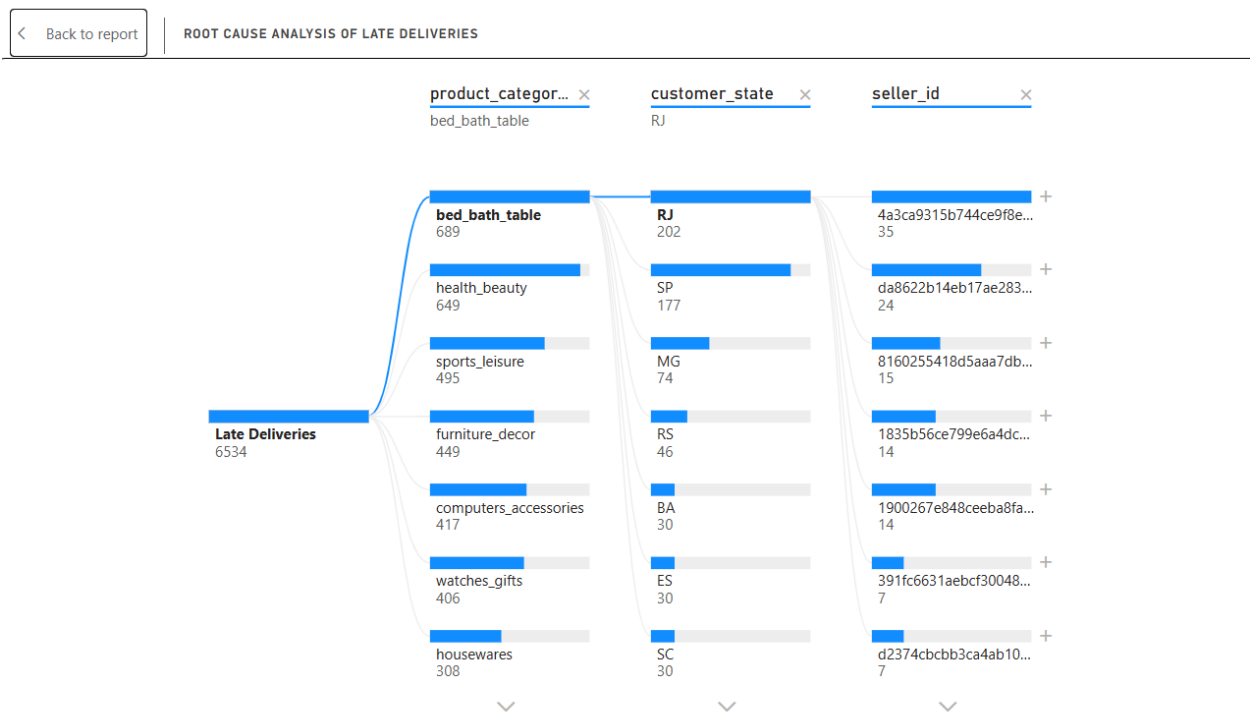


Figure 6: Root Cause Analysis of Late Deliveries

The decomposition tree in Figure 6 identifies the main contributors to SCGR’s late deliveries. At the product-category level, *bed_bath_table* accounts for the highest number of delays with 689 delays, followed by *health_beauty* (649) and *sports_leisure* (495). When expanded by customer state, regions such as *RJ* and *SP* emerge as significant contributors. The tree also reveals that a small number of sellers are disproportionately associated with delayed orders. This visualisation provides SCGR with actionable insights: managers can focus monitoring efforts on problematic categories, work with logistics partners in high-delay regions and intervene with specific underperforming sellers. This directly supports targeted performance improvement rather than broad, unfocused interventions.

3.6 Key Influencers of Customer Review Scores

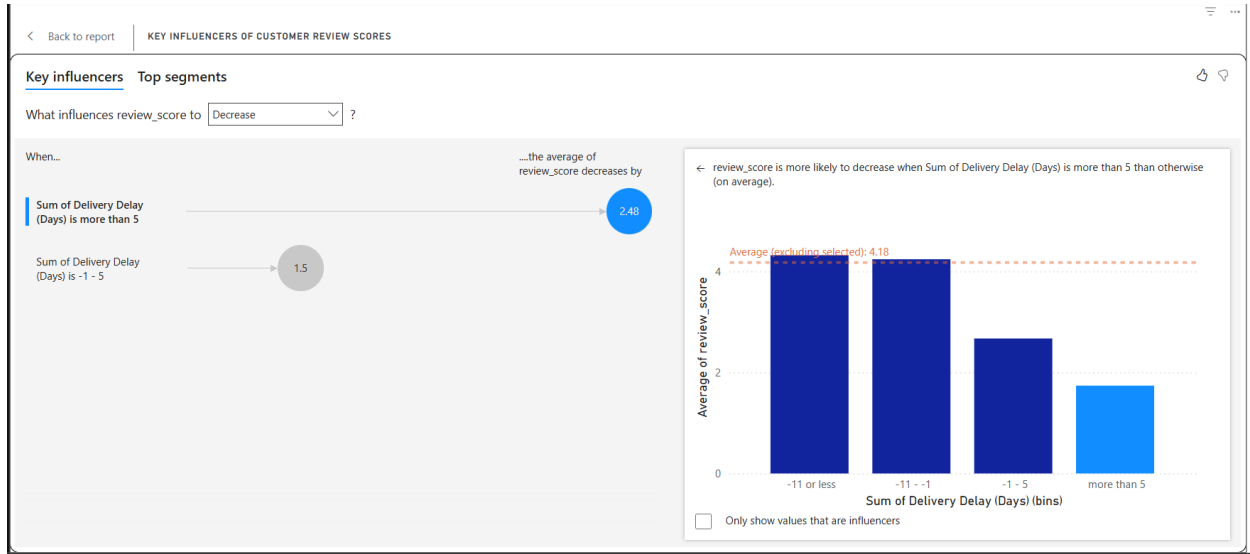


Figure 7: Key Influencers of Customer Review Scores

The Key Influencers visual (Figure 7) identifies the drivers that reduce customer review scores. The strongest negative factor is **delivery delay greater than 5 days**, which decreases the average review by **approximately 2.48 points**. Moderate delays (–1 to 5 days) also reduce satisfaction, though to a lesser extent. This confirms that delivery reliability is the primary determinant of customer experience on the SCGR platform. Although the positive influencer version of the analysis shows that early deliveries and some state-level factors can improve scores, the negative influencer model is more relevant to SCGR’s performance issue. Managers can use these insights to prioritise reducing delivery lead times, strengthening logistics partnerships and identifying categories most sensitive to delays.

3.7 Sales Contribution by Product Category (Treemap)

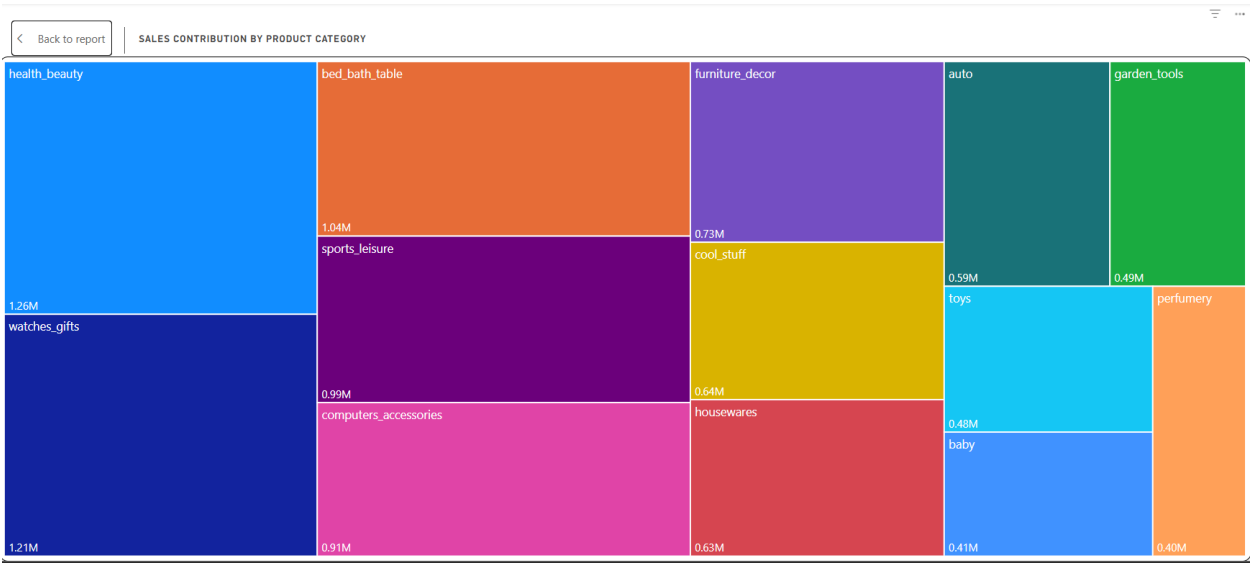


Figure 8: Sales Contribution by Product Category

The treemap in Figure 8 shows the distribution of total sales across product categories. Categories such as *health_beauty* (**1.26M**), *watches_gifts* (**1.21M**), *bed_bath_table* (**1.04M**) and *sports_leisure* (**0.99M**) contribute significantly to overall revenue. The benefit of this insight is that it allows SCGR to understand which categories drive the most value and may therefore require prioritised attention regarding stock management, marketing and logistics. When compared against the decomposition tree, the treemap also reveals mismatches between revenue and reliability; some high-revenue categories (e.g., *bed_bath_table*) also contribute heavily to delays. This supports more nuanced operational planning.

3.8 Relationship Between Delivery Delay and Review Score (Scatterplot)

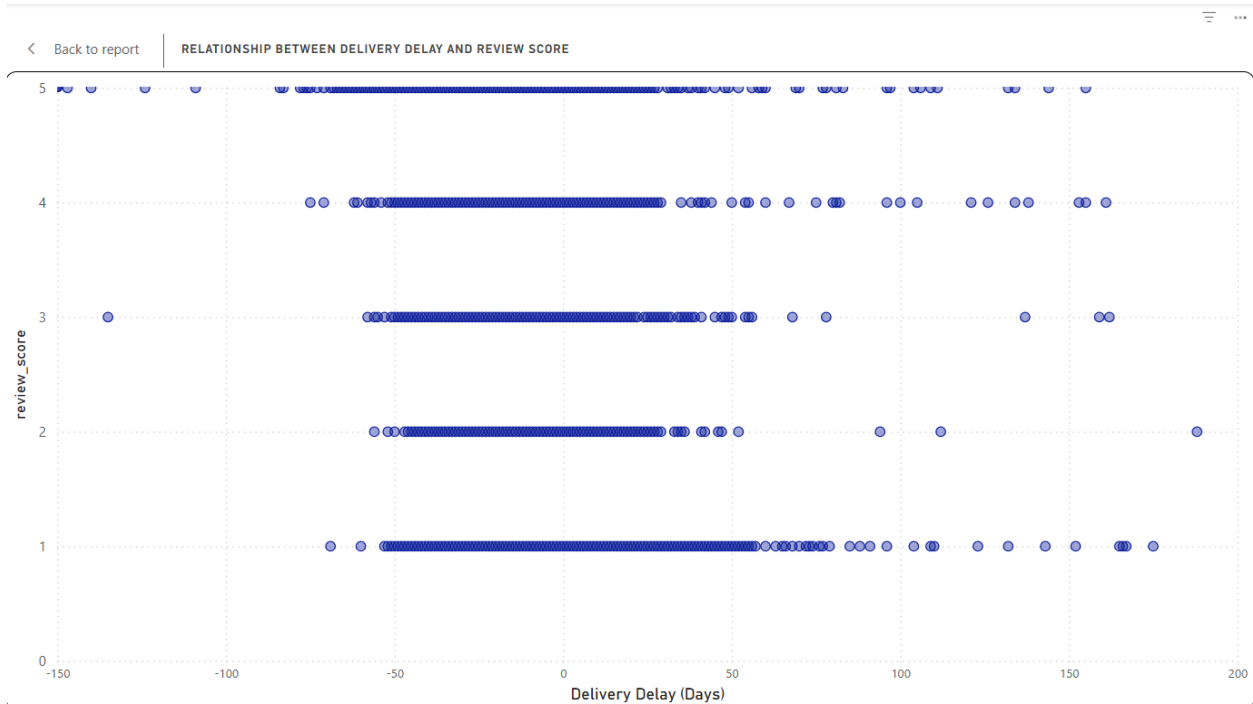


Figure 9: Relationship Between Delivery Delay and Review Score

Figure 9 shows the relationship between delivery delay and review score for individual orders. There is no strong linear pattern in the chart as 5-star reviews appear across almost the entire delay range, including many significantly late orders, while 1–3-star reviews are similarly dispersed. As review scores are integers, the horizontal groups are expected. Although extreme delay contributes to some low ratings, the overall distribution suggests that delivery timing alone does not fully explain customer satisfaction.

3.9 Geographic Distribution of Sales

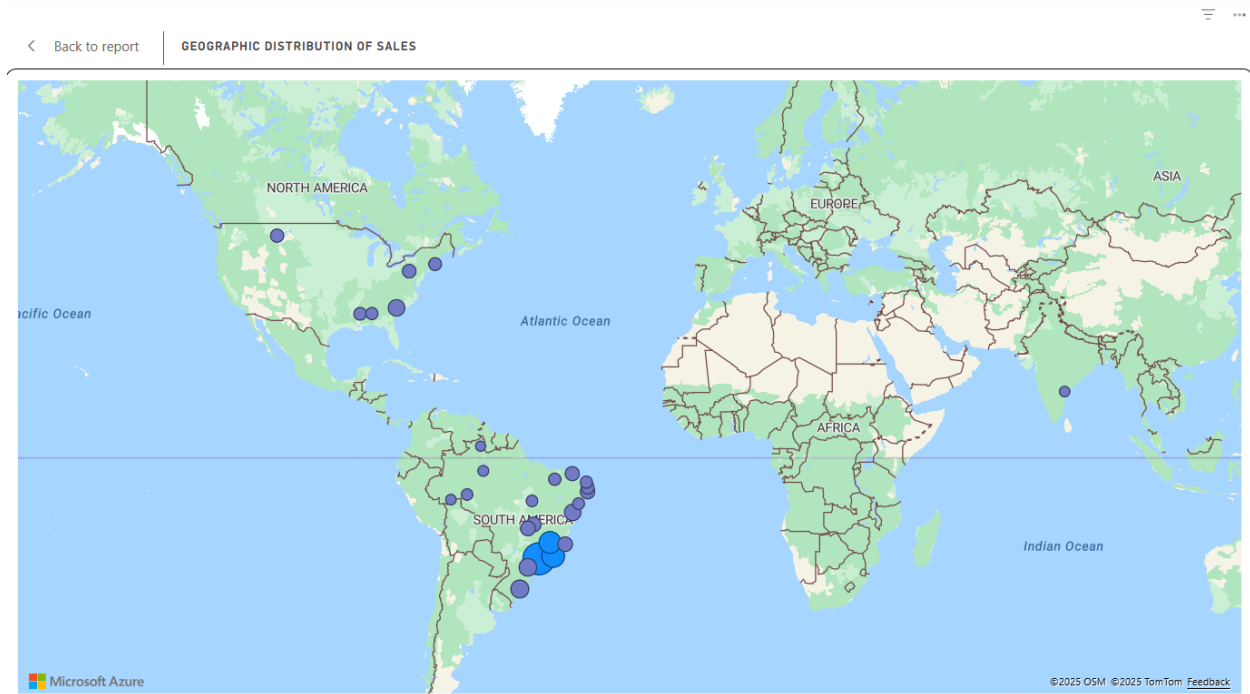


Figure 10: *Geographic Distribution of Sales*

Figure 10 shows that SCGR's sales are concentrated in Brazil, particularly in the South and South-East regions, where most demand and logistics activity occur. Several points appear outside Brazil; however, these are geocoding errors caused by Power BI misinterpreting Brazilian state abbreviations such as **SC, AL, PA, MA, MT and MS** as U.S. states (e.g., South Carolina, Alabama, Pennsylvania). These should therefore be interpreted as mapping errors rather than real customer locations. Despite this limitation, the map confirms SCGR's main operational footprint and helps managers understand regional demand patterns for routing and planning decisions.

3.10 Limitations of the Dashboard

Although the dashboard provides a comprehensive analytical overview, several limitations should be acknowledged. Several analyses can be influenced by occasional data quality issues, such as miscoded geolocations and inconsistent timestamps. Also, the absence of detailed logistics variables, such as warehouse locations, courier performance metrics or shipping

distances, limits the ability to fully analyse and diagnose delivery delays. The dashboard also relies on historical data only, meaning predictive elements (such as demand forecasting or delivery time estimation) are outside its current scope. Finally, customer review scores tend to be biased towards extremes, which may distort satisfaction patterns. Despite these limitations, the dashboard remains a valuable decision-support tool for SCGR and provides a strong foundation for further BI development.

4. Conclusion and Recommendations

The analysis conducted through SCGR's Business Intelligence dashboard provides a clearer understanding of the operational challenges affecting the organisation's performance, particularly in the areas of delivery reliability, customer satisfaction and category-level variation. The insights reveal that delivery delays and minor fluctuations in review scores point to areas where operational procedures need to be improved, even though overall sales and customer activity continue to grow. The dashboard shows that both logistics performance and product-specific factors shape customer experience. Overall, the BI approach strengthens SCGR's evidence-based decision-making. Although the scatterplot displays horizontal banding due to integer review scores, the density of lower ratings around longer delays reinforces that logistics performance remains a meaningful driver of customer dissatisfaction.

Based on these insights, several recommendations can be made to enhance SCGR's operational performance. First, SCGR should prioritise reducing delivery delays in high-impact categories such as *bed_bath_table* and *health_beauty*, as these contribute substantially to late deliveries, and delivery reliability is known to be a key driver of satisfaction and loyalty in e-commerce (Rashid and Rasheed, 2024). Regional delays could be addressed by strengthening logistics partnerships in key states, especially *RJ* and *SP*, since logistics service quality and last-mile performance are closely linked to perceived service quality (Nguyen et al., 2018). In addition, targeted monitoring of specific sellers associated with higher delay rates should be implemented, as addressing a small group of underperforming sellers may produce meaningful improvements in overall delivery reliability.

Second, the key influencer analysis demonstrates that delivery timing is a major determinant of review scores. Therefore, SCGR should invest in more accurate delivery promise windows and consider strategies that encourage earlier dispatch, because research shows deviations from promised delivery times have been shown to significantly affect satisfaction and evaluations of the retailer (Nguyen et al., 2018). To ensure expectations are being met, further evaluations of product quality, packaging standards, or seller practices may be necessary for categories where early delivery does not result in higher satisfaction, reflecting evidence that online reviews incorporate both experiential and expectation-based judgements (Filieri, 2015).

Finally, several enhancements could strengthen SCGR's long-term data capabilities and support deeper analysis by increasing geocoding accuracy, ensuring consistent timestamp recording and expanding the dataset to include logistics variables such as courier performance or warehouse information. Over time, SCGR could also incorporate predictive modelling to forecast demand, anticipate delivery bottlenecks and proactively manage seller performance, aligning with the established role of predictive analytics in supporting forward-looking decision-making and operational planning in Business Intelligence (Sharda et al., 2020). By implementing these recommendations, SCGR can build on the foundation established by the dashboard and move toward a more data-driven and operationally efficient business environment.

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